

Generation Interconnection System Impact Study Report

For Queue Project Q17-0796

Asheville 138 kV

January 19, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0796. The Interconnection Customer proposes a 43.9 MW battery energy storage facility interconnecting at the Asheville 138 kV substation in the Virginia area. The request was received on May 09, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 280 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,663,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Mcclain Energy Facility - Coal Substation 138kV line	Manatee - Hinton Ap 345kV	110.3	254	6.0
Beardsley St - Red Hills Substation 138kV	Hinton Ap - Lowmoor 345kV	104.6	355	10.5
Hinton Ap - Onard 138kV	Mcclain Energy Facility - Coal 138kV	103.9	229	8.4
Candelaria - Reach 138kV L	Bear - Clover 345kV	103.9	547	5.1
De Kalb - Oxnard 138kV	Coal - Mcclain Energy Facility 138kV L	103.8	223	8.4
New London - Declo 138kV	Bath County - Sherburne 345kV	102.9	307	5.8
De Kalb - Mcclain Energy Facility 138kV	Mcclain Energy Facility - Coal 138kV	102.4	220	8.4
Roanoke Rapids Substation - Mccartney 345kV line	Mccartney - Gaston 345kV L	101.3	1,852	7.3
Mccartney - Gaston 345kV	Mccartney - Roanoke Rapids 345kV	101.1	1,659	7.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mcclain Energy Facility - Coal Substation 138kV line	Rebuild line with higher-capacity conductor	\$6,381,000	4% (\$230,000)
Beardsley St - Red Hills Substation 138kV L	Reconductor line	\$5,673,000	1% (\$68,000)
Hinton Ap - Onard 138kV	Reconductor line	\$1,411,000	4% (\$55,000)
Candelaria - Reach 138kV L	Reconductor line	\$3,811,000	1% (\$28,000)
De Kalb - Oxnard 138kV	Reconductor line	\$2,099,000	4% (\$82,000)
New London - Declo 138kV	Reconductor line	\$21,974,000	1% (\$174,000)
De Kalb - Mcclain Energy Facility 138kV	Reconductor line	\$2,944,000	4% (\$115,000)
Mccartney - Roanoke Rapids 345kV	Reconductor line	\$24,347,000	2% (\$466,000)
Mccartney - Gaston 345kV	Reconductor line	\$23,240,000	2% (\$445,000)
	Total allocated to project		\$1,663,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.